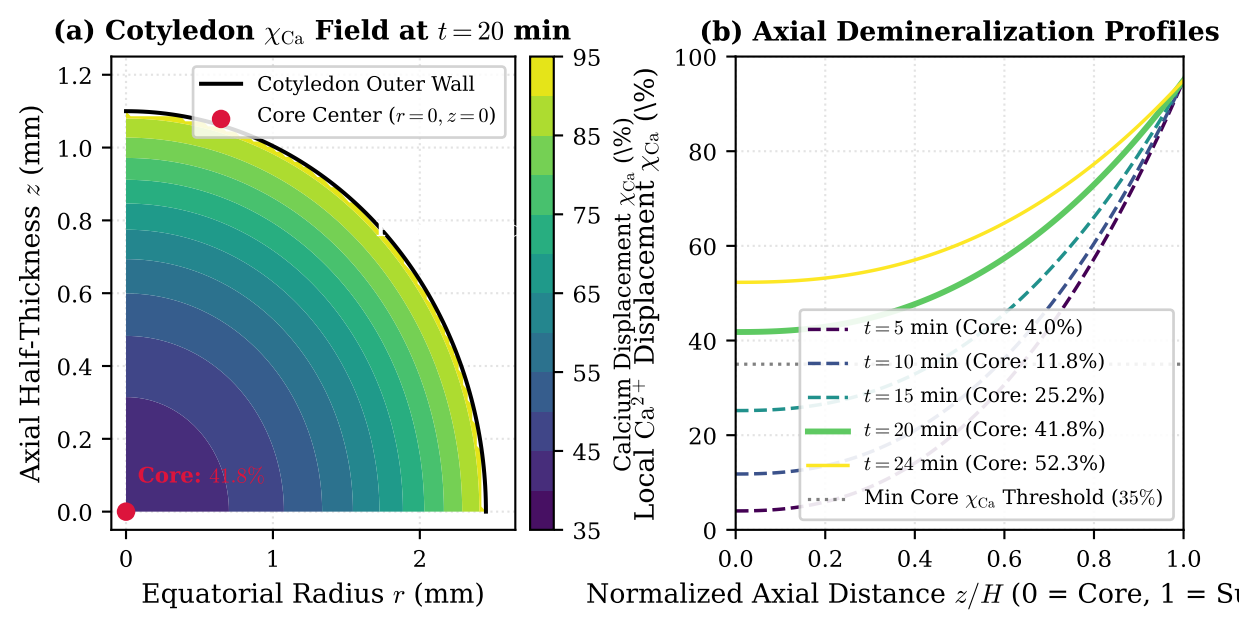

(a) Cotyledon χ_{Ca} Field at $t = 20$ min
Cotyledon Outer Wall
Core Center ($r = 0, z = 0$)
Core: 41.8%
Axial Half-Thickness z (mm)
Equatorial Radius r (mm)
Calcium Displacement χ_{Ca} (%)
Local Ca^{2+} Displacement χ_{Ca} (%)
(b) Axial Demineralization Profiles
$t = 5$ min (Core: 4.0%)
$t = 10$ min (Core: 11.8%)
$t = 15$ min (Core: 25.2%)
$t = 20$ min (Core: 41.8%)
$t = 24$ min (Core: 52.3%)
Min Core χ_{Ca} Threshold (35%)
Normalized Axial Distance z/H (0 = Core, 1 = Su